Technology and Health Care 32 (2024) 4965–4982
DOI 10.3233/THC-240158
IOS Press

CNN-based glioma detection in MRI: A deep learning approach

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Received 20 January 2024
Accepted 13 March 2024

Abstract.
BACKGROUND: More than a million people are affected by brain tumors each year; high-grade gliomas (HGGs) and low-grade gliomas (LGGs) present serious diagnostic and treatment hurdles, resulting in shortened life expectancies. Glioma segmentation is still a significant difficulty in clinical settings, despite improvements in Magnetic Resonance Imaging (MRI) and diagnostic tools. Convolutional neural networks (CNNs) have seen recent advancements that offer promise for increasing segmentation accuracy, addressing the pressing need for improved diagnostic and therapeutic approaches.
OBJECTIVE: The study intended to develop an automated glioma segmentation algorithm using CNN to accurately identify tumor components in MRI images. The goal was to match the accuracy of experienced radiologists with commercial instruments, hence improving diagnostic precision and quantification.
METHODS: 285 MRI scans of high-grade gliomas (HGGs) and low-grade gliomas (LGGs) were analyzed in the study. T1-weighted sequences were utilised for segmentation both pre-and post-contrast agent administration, along with T2-weighted sequences (with and without Fluid Attenuation by Inversion Recovery [FAIRE]). The segmentation performance was assessed with a U-Net network, renowned for its efficacy in medical image segmentation. DICE coefficients were computed for the tumour core with contrast enhancement, the entire tumour, and the tumour nucleus without contrast enhancement.
RESULTS: The U-Net network produced DICE values of 0.7331 for the tumour core with contrast enhancement, 0.8624 for the total tumour, and 0.7267 for the tumour nucleus without contrast enhancement. The results align with previous studies, demonstrating segmentation accuracy on par with professional radiologists and commercially accessible segmentation tools.
CONCLUSION: The study developed a CNN-based automated segmentation system for gliomas, achieving high accuracy in recognising glioma components in MRI images. The results confirm the ability of CNNs to enhance the accuracy of brain tumour diagnoses, suggesting a promising avenue for future research in medical imaging and diagnostics. This advancement is expected to improve diagnostic processes for clinicians and patients by providing more precise and quantitative results.

Keywords: Biomedical imaging, CNN, deep learning, glioma, neuroimages, MRI

1. Introduction

Biomedical imaging techniques involve the acquisition, processing, and subsequent analysis of two-or three-dimensional images that visually represent the internal parts of the human or animal body. These images are obtained from medical imaging devices such as magnetic resonance imaging (MRI) machines, computed tomography (CT), positron emission tomography (PET), ultrasound, and so forth [1,2]. This

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field provides a plethora of invaluable tools for clinical analysis, enabling the assessment of organ and biological system functions, as well as aiding in the diagnosis and monitoring of disease treatment. Often, multiple imaging modalities are employed concurrently to maximize the amount of information derived from a patient [3,4,5,6,7]. The domain of medical image processing utilizes mathematical tools, including spatial, temporal, and spectral analysis, stochastic methods, and machine learning, to develop software [8]. This software aids in addressing medical challenges, such as the early and differential diagnosis of disorders, and in developing personalized and precision medicine therapies for various diseases. Recently, MRI techniques have experienced significant advancements in both the technology of devices and diagnostic algorithms, particularly in diagnosing disorders at the cellular level [9,10,11,12].

Brain tumors, also known as intracranial neoplasms, are pathological conditions characterized by the unchecked growth of abnormal cells within the brain. The American Cancer Society reports that the survival rates for tumors in the central nervous system are alarmingly low, ranging between 27% to 33%, depending on the tumor type and stage of advancement [13,14]. The most prevalent form of brain cancer is glioma, which originates in the glial tissue [15]. Malignant tumors typically arise from glial tissue, comprised of glia cells responsible for providing mechanical support to the brain's nerve cells. Gliomas represent the most severe form of brain tumors, accounting for 78% of malignant brain tumors and standing as the most common type among adults [16]. They often lead to cerebral edema, which involves swelling due to unchecked fluid accumulation in the brain. This condition exacerbates the mortality risk associated with gliomas, as the accumulated fluids facilitate the invasion of cancerous glial cells. Furthermore, the swelling from edema physically compromises brain function by compressing and displacing nearby nerve cells, potentially leading to their dysfunction [17].

Machine learning, a field dedicated to developing iterative and adaptive computational algorithms through the analysis of large datasets, has yielded remarkable outcomes across various applications, including pattern recognition, natural language processing, data classification, and image segmentation [18]. The advent of powerful graphics processing units (GPUs) has significantly enhanced the training of deep neural networks, a subfield known as deep learning. This advancement has positively impacted algorithm performance, especially in applications using convolutional neural networks (CNNs) across diverse areas. These areas range from pathology detection in medical images and autonomous vehicle control to automatic language translation and the discovery and toxicological analysis of new drugs [19]. The field's state of the art is continuously evolving, as evidenced by participation in major international challenges like the Multimodal Brain Tumor Image Segmentation Benchmark (BRATS) [20] and the ImageNet Large Scale Visual Recognition Challenge [21,22]. In medical imaging, CNNs have been successfully applied to address segmentation and classification challenges, achieving outstanding results thanks to the expansion of large-scale medical image databases. Current magnetic resonance images, known for their rich detail and soft tissue information, benefit from signal and image processing techniques such as filtering, segmentation, and feature extraction. These techniques are commonly utilized in computer-aided diagnostic (CAD) systems to enhance diagnostic accuracy and efficiency [11].

After acquiring and reconstructing medical images in the digital domain, the images come to exist virtually as two-dimensional matrices. Each position of the matrix element corresponds to the positions of the image pixels, and the values of each element correspond to the intensity of the light signal or visual information that has been digitized. This study uses concepts such as image segmentation, data normalization and machine learning to segment cancerous regions and tissue.

The paper describes a study that used deep convolutional neural networks (CNNs) to rapidly classify and separate gliomas in multimodal magnetic resonance images. It provides an automated segmentation strategy for gliomas and their components, with segmentation performance equivalent to experienced

Table 1
A comparative analysis for many previous studies

Study	Authors	Method	Dataset	Main findings	Challenges addressed
[30]	Saeedi et al., 2023	2D CNN, Convolutional Auto-Encoder	3264 MRI Images	Proposed 2D CNN achieved 96.47% accuracy, Auto-Encoder 95.63%	Early detection of brain tumors
[31]	Tiwari et al. 2023	Efficient-NetB0, VGG16, ResNet50, InceptionV3, MobileNetV2	Benchmark data	Novel data augmentation with state-of-the-art models	Complex MRI diagnosis process
[32]	Ahmed et al., 2022	Modified ResNet50	3586 T1-w MRI Images	Achieved highest accuracy of 99.8%	Accurate brain tumor classification
[33]	Shomirov et al., 2022	WFL-Based 3D U-Net	BraTS 2019, 2020	Improved segmentation for HGG and LGG	Class imbalance in segmentation
[34]	Aboelenein et al., 2022	MIRAU-Net	Brats 2019	Mean dice similarities: 0.885 WT, 0.879 TC, 0.818 ET	Class imbalance and severity variation in gliomas
[35]	Wang et al., 2021	TransBTS	BraTS 2019, 2020	Achieved comparable or higher results than previous methods	Integrating local and global features
[36]	Sheng et al., 2021	SoResU-Net	BraTS 2018, 2019	Outperforms baseline, especially on core and enhancing tumor segmentation	Utilizing second-order statistics for segmentation

radiologists and commercial technologies. The methodology consists of a literature review, data extraction from the BRATS 2016 database, preprocessing, neural network training, and analysis. Two neural network designs were created for multiclass segmentation and anomaly segmentation, and they produced promising results despite the necessity for image resolution reduction. The study implies that computational strategies based on machine learning and signal processing can effectively handle complicated medical and biological issues.

1.1. Literature review

Table 1 shows a comparative analysis for many previous studies:

1. Differences
   - Approaches: The methods vary from basic 2D CNNs and convolutional auto-encoders to advanced structures like EfficientNetB0, VGG16, ResNet50, InceptionV3, MobileNetV2, and unique models such as customized ResNet50, WFL-Based 3D U-Net, MIRAU-Net, TransBTS, and SoResU-Net. This diversity illustrates the changing environment of deep learning in medical imaging, utilizing both 2D and 3D models to enhance accuracy and efficiency.
   - Datasets: The studies use various datasets, such as benchmark data and particular datasets like BraTS (2019, 2020, 2018), as well as custom collections of MRI images. The selection of dataset influences the model's training, as larger and more diverse datasets usually result in more resilient models.
2. Improvements
   - Accuracy: Most studies indicate a high level of accuracy in tumor detection and segmentation, with some obtaining up to 99.8% accuracy. The advancements are attributed to novel neural network structures and sophisticated training methods.

- Segmentation Performance: Methods such as WFL-Based 3D U-Net and MIRAU-Net have demonstrated notable enhancements in accurately segmenting high-grade gliomas (HGG) and low-grade gliomas (LGG), as well as identifying tumor core (TC) and enhancing tumor (ET) regions.
- Handling Class Imbalance and Complexity: Various studies have proposed techniques to tackle class imbalance, as demonstrated in the works of Shomirov et al. and Aboelenein et al., and to address the inherent complexity of MRI diagnosis, as discussed by Tiwari et al. and Wang et al. These efforts highlight the continuous work being done to enhance deep learning models for practical medical imaging tasks.

3. Limitations
   - Generalization: Although these models exhibit great accuracy and enhanced segmentation performance, their capacity to generalize across various datasets and real-world situations still poses a difficulty. Performance may differ dramatically when models are used with data from various sources or imaging procedures.
   - Computational Resources: Advanced models, particularly those using 3D CNNs and Transformer-based architectures, demand significant computational resources for training and inference, thus restricting their usability and feasibility in certain clinical environments.
   - Interpretability: Deep learning models are often intricate, resulting in a "black box" issue, making it difficult to understand the decision-making process of these models. This is a significant constraint in medical applications, as comprehending the reasoning behind a diagnosis or segmentation is essential for establishing confidence and dependability.

2. Methodology

In this section, the methodology used for the development of a brain tumor segmentation algorithm in magnetic resonance images is presented, as proposed in this course conclusion work. Figure 1 represents a summary of the process implemented in this study.

The methodology used in this study was as follows:

- Literature review on the current challenges encountered in modern medical imaging;
- Bibliographic review of the techniques used to solve the challenges and the available open data sets;
- Extraction of data from the BRATS 2016 database;
- Study of different neural network architectures used in medical segmentation tasks;
- Preprocessing of the dataset to make it possible to train the network in practical time;
- Division of the preprocessed set into two subsets: training and testing
- Neural network training and acquisition of results
- Analysis of results and preparation the report.

The program was completely developed using Python 2.7, with the help of open-source libraries such as VTK, SimpleITK and OpenCV for data preprocessing, KERAS for neural network development, Numpy and Matplotlib for data visualization and organization, plus Jupyter for software development and interactive debugging.

2.1. Challenges in CNN model training

Within our methodology, training a Convolutional Neural Network (CNN) for accurate glioma segmentation in MRI images faced distinct problems. These difficulties originate from the inherent diversity

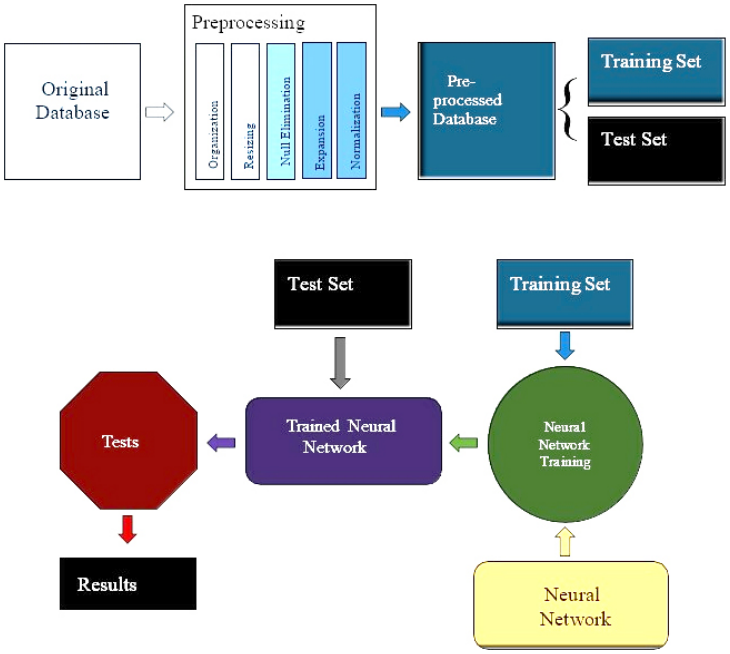


Fig. 1. Processing and training scheme implemented.

of tumor appearance among patients, the large imbalance between tumor and non-tumor regions within the datasets, and the high demand for adequately annotated training data.

- Variability in Tumor Appearance: The size, form, and location of gliomas vary greatly between people, adding significant difficulty. This variability needs a model that can generalize from highly heterogeneous data to enable correct segmentation across the glioma spectrum.
- Imbalance Between Tumor and Non-Tumor Regions: MRI scans include a large number of non-tumor regions by definition. This imbalance raises the possibility of the model having a bias toward predicting non-tumor areas, potentially missing smaller or less identifiable tumor spots. Addressing this imbalance was critical for improving our model's sensitivity to glioma characteristics.
- Need for Annotated Training Data: Creating a strong CNN model requires a substantial dataset of accurately labeled pictures. Compiling a dataset for gliomas that is both large and accurately annotated is a challenging task due to their delicate and complicated nature. The accuracy of our model's segmentation skills is closely linked to the quality and quantity of the training data provided.

To overcome these obstacles, we adopted a variety of solutions. Data augmentation approaches were used to increase diversity in the training set, imitating a wider range of tumor appearances. In addition, we investigated various techniques for addressing class imbalance, such as using weighted loss functions that prioritize the minority class. Our dataset was meticulously curated and annotated by clinical specialists, resulting in high levels of accuracy in our training data.

The U-Net architecture was employed for segmentation, chosen for its efficiency in medical image segmentation tasks, offering significant improvements in Dice coefficient values for NR, total tumor, and tumor nucleus.

Model Robustness and Overfitting: The study utilized a varied and extensive dataset from BRATS 2016, incorporated dropout and batch normalization methods in the CNN design to avoid overfitting, and divided the dataset into training and test sets for unbiased model evaluation.

Table 2
Sizes of data tensors once imported into the python environment

Data	Size
MRI images	$274 \times 4 \times 155 \times 240 \times 240$
Segmentation masks	$274 \times 5 \times 155 \times 240 \times 240$

2.2. BRATS 2016 database

BRATS database was created to facilitate the validation and comparison between studies and brain tumor segmentation strategies based on multimodal image data obtained from several simultaneous medical imaging modalities [20]. The BRATS 2016 edition features datasets obtained in 2012 and 2013 by BRATS itself and images provided by the NIH Cancer Imaging Archive, a virtual and global medical imaging center [22,23].

This database was chosen for several reasons. First, all images of the different modalities are recorded per patient; that is, brain images obtained from each patient are aligned without the need to study alignment methods, which would consume time and deviate from the central theme of this study. In addition, the database contains images from four MRI modalities: T2 FLAIR, T2, T1 with contrast and T1. All images obtained during the same imaging sessions for each patient were interpolated to 1 mm^3 voxel resolution and manually segmented by up to four medical professionals. Segmentations were performed to separate the tumor nucleus, the region directly affected by the tumor, the necrotic (dead) region, and the region affected by edema.

2.2.1. Data structure

The database is organized into folders separated initially by tumor grade, low grade (1 to 2, benign), or high grade (3 to 4, malignant). Data from 220 subjects with malignant tumors and 54 with benign tumors were obtained and used in this study. For each individual, the image data for each modality is separated into its subfolder. The images have a 240 by 240 pixels resolution, and there are 155 images (slices, z-axis) per modality. There are 4 modes in total. Allied to the MRI images are then segmented masks for each characteristic: cancer nucleus, cancer affected region, dead region, edematous region and the last mask marking only the healthy region or not affected by edema or cancer [24]. These masks are in only one image per slice, with each character corresponding to a specific value: 1, 2, 3, 4 and 5. A simple algorithm was created to separate these masks into 5 images, one more to identify the region not affected by tumor or edema. Figure 2 illustrates an example of this resulting bank.

2.3. Pre-processing

In this section, the data processing steps to use in the neural network training will be treated efficiently [23,25].

2.3.1. Organization

The first step is the organization of the database. All data were imported into the Python environment and organized in the form of tensors, or multidimensional matrices, of sizes (Table 2).

In this study, the data were treated as bidimensional; the tensor was reformatted to stack all the patients' slices in a single pile, eliminating one of the tensor axes. The size of the resulting tensors (Table 3).

In this way, it is possible to access any image from a numerical specification on the first axis; for example, accessing element 50 would be given as follows: matrix [50], returning a matrix $k \times 240 \times 240$ ($k = 4$ for the MRI images and $k = 5$ for the segmentation masks), similar to what would occur in an RGB image ($k = 3$ matrices, one for each color channel).

Table 3
Tensioner sizes after stacking

Data	Size
MRI images	$42470 \times 4 \times 240 \times 240$
Segmentation masks	$42470 \times 5 \times 240 \times 240$

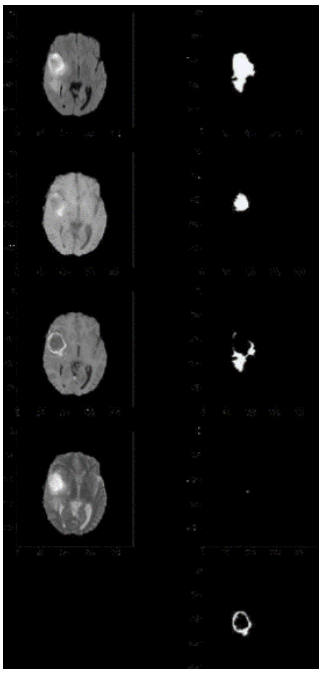


Fig. 2. Sample BRATS 2016 sample from a single patient. (Source: BRATS 2016 database).

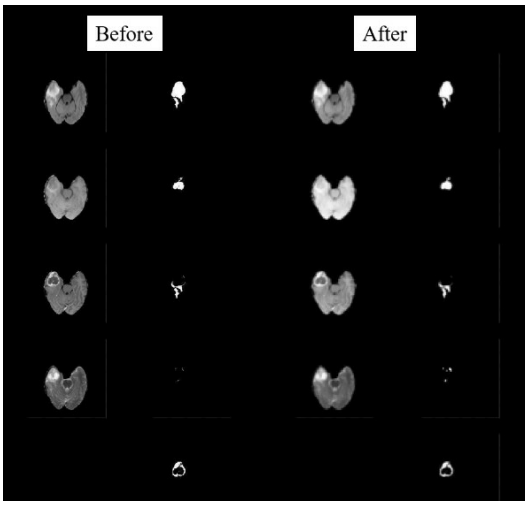


Fig. 3. Sample resized to 60×60. (Source: BRATS 2016 database).

Table 4
Tensor sizes after resizing

Data	Size
MRI images	$42470 \times 4 \times 60 \times 60$
Segmentation masks	$42470 \times 5 \times 60 \times 60$

2.3.2. Resizing

In the second step, all images were resized from 240×240 resolution to 60×60, that is, a 4x sub-sizing. The algorithm for the images' decimation was the bilinear decimation, as it is fast and does not cause high-frequency artefacts as in the nearest neighbor decimation. Figure 3 shows an example image after resizing. Note that both the masks and the MRI images have been resized.

This process was implemented to make the practical training of the neural network feasible. Despite its benefits, it is undeniable that the resizing process can diminish the neural network's performance due to the loss of data. Without this resizing, training the neural network could take up to 5 days on the computer utilized for these experiments, as determined by preliminary tests. By reducing the image sizes, the training duration was shortened to approximately 8 hours. This significant reduction in time enables the execution of multiple experiments with various neural network architectures. Table 4 presents the characteristics of the dataset after the resizing process.

In the mask images, which originally only contained values 0 and 1, the bilinear decimation process generated intermediate values between 0 and 1, which could be rounded, generating the equivalent of a neighbor nearest decimation. This was not done because the intermediate values can be useful for training the neural network since there is information contained in them, and eliminating them would be equivalent to eliminating this information.

2.3.3. Elimination of null elements

The third preprocessing step involved eliminating excess null elements in the data. The database, as it comprises MRI scans of the human brain along a vertical axis, has a number of images in which there is no segmentation mask, as there is no cancer in these images. Figure 4 shows an example of a null image [26].

This step serves to homogenize the database, preventing the neural network from being trained with data mostly of only one type, that is, without a segmentation mask. This would lengthily be training and generate null local minima in training the network.

The elimination algorithm randomly discovers images whose mask values are all null in the database. A count is performed to record the number of elements and which elements are null, and a predefined percentage of these elements is eliminated.

It's crucial to highlight that when training neural networks, employing a dataset that closely mirrors real-world scenarios is considered best practice. In practical applications, many of the images the network will encounter may not display any anomalies, such as cancer or edema. This implies that completely eliminating all images lacking cancerous conditions is not advisable; instead, only a portion should be removed. For experimental purposes, an arbitrary elimination percentage was established. Approximately 50% of the non-anomalous (null) data was chosen to be removed. This percentage was maintained consistently throughout the training process. Altering this value to enhance network performance could inadvertently lead to network overfitting, a scenario where the model performs well on the training data but poorly on unseen data.

The null images comprise about 50% of the total database; that is to say, the final database has approximately 50% (non-null images) + 50% $\times$ 50% (a non-eliminated portion of the null images) =

Table 5
Tensor sizes after resizing

Data	Size
MRI images	$31852 \times 4 \times 60 \times 60$
Segmentation masks	$31852 \times 5 \times 60 \times 60$

Table 6
Characteristics of the dataset after preprocessing [27]

Data	Size	Range of values
MRI images	$31852 \times 4 \times 60 \times 60$	0 to 1
Segmentation masks	$31852 \times 5 \times 60 \times 60$	0 to 1

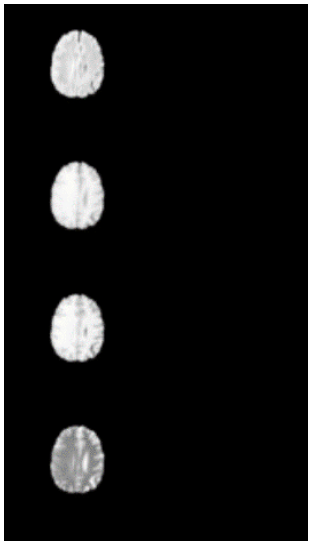


Fig. 4. Example of a null sample, used in the study.

75% of the original data size. As the algorithm randomly decides with a 50% probability whether each null image is eliminated or not, the final number of samples in the database varies each time the program is run. Table 5 shows an average estimate of the size of the database after removing the null data.

2.3.4. Standardization

The subsequent phase in data preprocessing involves normalization. This step standardizes the scale and mean value of the data, ensuring that the values input into the neural network's activation functions fall within a consistent range. This uniformity is crucial for the neural network to process and learn from the data effectively. For MRI images, normalization was achieved by scaling the grayscale pixel values, which originally range from 0 to 255, to a range of 0 to 1. This scaling was accomplished by dividing all the grayscale values by 256. The mask images, used to identify specific areas of interest within the images, required no modification as their values already fell between 0 and 1.

2.3.5. Training and test set

The resulting dataset has the characteristics represented in Table 6.

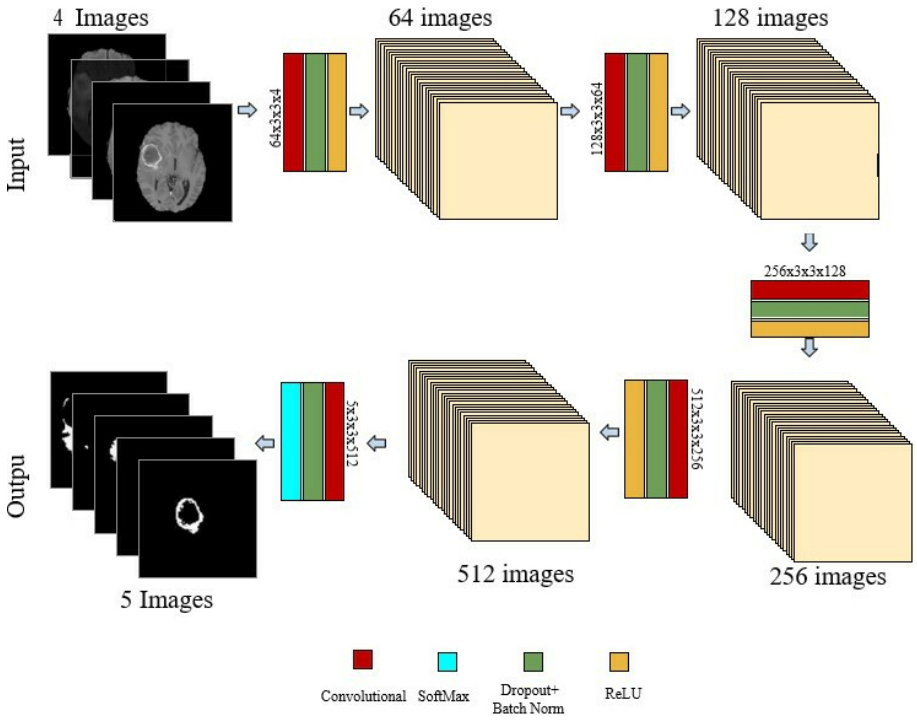


Fig. 5. Architecture A: Multiclass segmentation.

2.4. *Proposed architectures of neural networks*

2.4.1. *Architecture A: Multiclass segmentation*

The proposed CNN architecture [28] has 5 convolutional layers, with 3×3 convolutional cores. The layers have increasing numbers of filters with each layer, doubling the number of filters in the previous layer. Where, the dataset consisted of 285 MRI pictures of high-grade and low-grade gliomas, obtained from the BRATS 2016 database. The photos consist of a variety of patient demographics, with information from 220 individuals with malignant tumors and 54 with benign tumors. The photographs encompass different MRI modalities and have been categorized by medical experts. The first layer has 64 filters as a base, and subsequent layers double the number of filters from the previous layer. In the fifth layer, the output layer, only 5 filters are used to join and filter information from the 512 outputs of the previous layer, joining them into 5 final output images, each representing one of the masks of each output class. The architecture is illustrated in Fig. 5. This scheme should encourage learning of basic filters, such as edge detection, smoothing, and contrast enhancement, in the initial layers. In contrast, subsequent layers can use images from previous layers to detect more enhanced features or complexity. All convolutional layers are accompanied by batch normalization and dropout.

2.5. *Architecture B: Anomaly segmentation*

For a simpler task, a secondary architecture was designed to use only the first image of the BRATS set. This image contains pixels with a value equal to 0 for those corresponding to the positions in the original image that do not have a brain anomaly, such as edema or cancer, and pixels equal to 1 for those that

Table 7
Characteristics of the training set used in the neural network

Data	Size	Range of values
Entries (MRI images)	$22296 \times 4 \times 60 \times 60$	0 to 1
Targets (targeting masks)	$22296 \times 5 \times 60 \times 60$	0 to 1

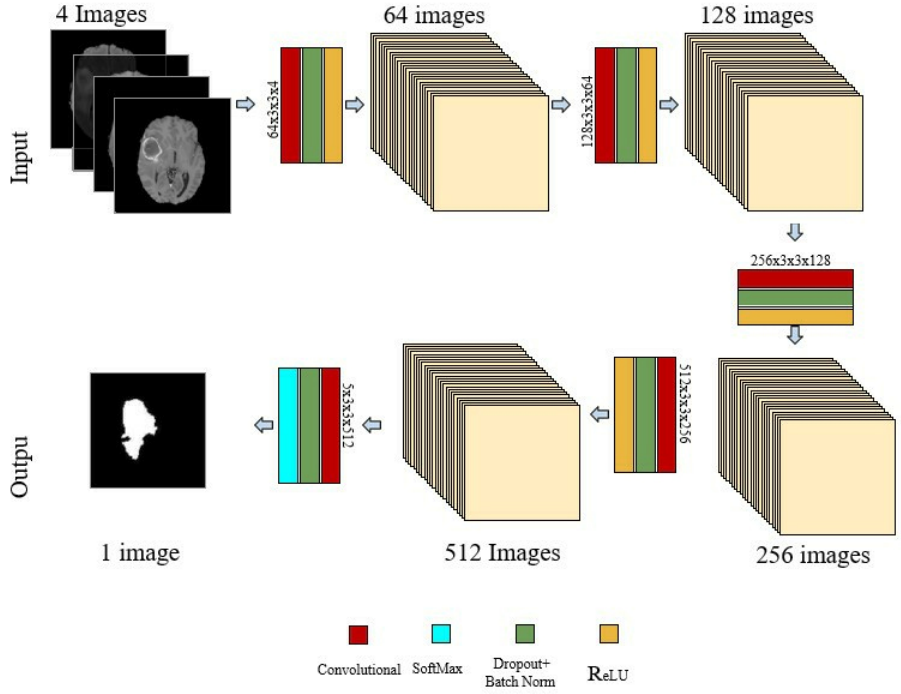


Fig. 6. Architecture B: Anomaly segmentation.

contain these anomalies. In this way, the neural network can be trained only to recognize brain damage or anomaly without distinguishing internal details of the anomaly, as proposed in the previous architecture. This reduces the complexity of the task to be learned by the neural network, allowing greater precision and less training time. Figure 6 illustrates the proposed architecture, which differs only in the number of output images.

2.6. Neural network training

2.6.1. Separation of the data set

For training the neural network, the resulting dataset was separated into the test set and the training set [29]. The distribution was done randomly so that 70% of the data went to the training set and the other 30% to the test set. Thus, the neural network can be trained with one set and tested with a different set to obtain performance metrics without bias (Table 7 and Table 8).

2.6.2. Tests performed

The experiments conducted were premised on the understanding that each loss function yields varying outcomes in networks tasked with different operations. Consequently, two distinct architectural models,

Table 8
Characteristics of the test set used in the neural network

Data	Size	Range of values
Entries (MRI images)	$9557 \times 4 \times 60 \times 60$	0 to 1
Targets (targeting masks)	$9557 \times 5 \times 60 \times 60$	0 to 1

referred to as Architecture A and B, were evaluated using two loss functions highlighted in this research: the Dice score and cross-entropy.

Architecture A underwent training utilizing the cross-entropy loss function, chosen for its applicability in multiclass segmentation tasks. The cross-entropy parameter is beneficial because it can differentiate between multiple classes, an essential feature for the tasks assigned to Architecture A. On the other hand, Architecture B was trained using the Dice score approach, selected due to its suitability for binary classification problems. The Dice score, in its conventional form, is limited by its inability to distinguish between various classes, rendering it ideal for binary classification scenarios only.

To assess the outcomes, segmentation masks produced by each model were evaluated and compared against ideal masks to calculate the Dice score for each result. This analysis was conducted for both architectures, irrespective of the loss function employed during their training. Thus, even for the network trained with cross-entropy, the Dice score was utilized as a quantitative metric to gauge the performance of the segmentation algorithm. Employing the Dice score as a common measure across both architectures facilitates a more objective comparison of their results.

3. Results

The neural network architectures were trained over 3 weeks. The computer used for training had a Core i7 processor (3.3 GHz) with four cores, 8 GB of on-board 1600 MHz RAM and an SSD-type disk.

After training the proposed networks, the test sets were applied to the networks' inputs and, with each sample's desired outputs, the Dice scores were calculated. For each architecture, the means and standard deviation of all Dice scores obtained were calculated as an average value $43.33 \pm 11.94\%$ in architecture A and for architecture B an average value $89.61 \pm 18.22\%$.

We can see examples of images from training the architecture A: multiclass neural network. The images in Fig. 7 are examples of the best results obtained with the training of this network. It can be seen that the network was able to learn not only about the affected area but also more abstract characteristics of the affected region, being able to discern between different types of tissue and damage, such as oedema and the central region of the tumor. Figure 7 (Example 4) shows examples of less successful segmentation, which generally involve the outermost parts of the brain, where there is less information in the training set because tumors generally develop closer to the innermost region of the brain. brain. In addition, these images show a weak point in the network, which is to discern between the higher contrast details of the edges of the brain, and brain tumors or anomalies.

3.1. Segmentation performance across glioma grades

Understanding the differences between high-grade gliomas (HGGs) and low-grade gliomas (LGGs) is critical for accurate treatment planning and prognosis. This investigation seeks to illuminate our model's segmentation performance across various categories, providing key insights for improving diagnostic precision. We divided our dataset into HGGs and LGGs using a methodological approach inspired by the

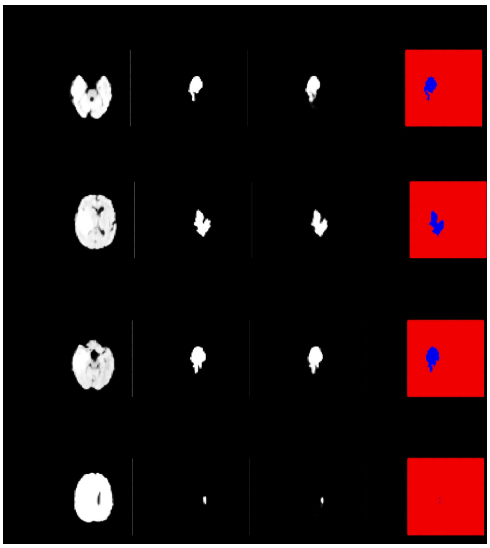


Fig. 7. Examples of architecture A test result.

"Evolutionary Model for Brain Cancer-Grading and Classification" (IEEE Access), which provided a solid foundation for our investigation.

Our findings reveal distinct Dice coefficients, sensitivity, and specificity metrics for HGGs and LGGs. The model showed a greater Dice coefficient in HGGs, perhaps because of their more prominent imaging features compared to LGGs. This difference highlights how the model is affected by tumor diversity and the significance of imaging features in glioma segmentation. Our model's competitive advantage in accurately segmenting gliomas across various grades is underscored by comparing it with the IEEE Access study and other relevant research.

Comparison of Segmentation Accuracy between High-Grade and Low-Grade Gliomas: The study does not clearly distinguish the segmentation performance for high-grade gliomas (HGGs) against low-grade gliomas (LGGs). Future updates of the paper should analyze model performance separately for HGGs and LGGs due to their distinct treatment paths and prognosis. One approach is to compare the Dice coefficient, sensitivity, and specificity of segmentation results between high-grade gliomas (HGGs) and low-grade gliomas (LGGs), maybe utilizing information from the IEEE Access publication for analysis and discussion.

Appendix D, in turn, has examples of training results from architecture B. These examples, particularly those in Fig. 8, show how the network trained by optimizing the Dice score in the single-class problem can obtain more robust results than the multiclass problem of architecture B. The images in example 4 of both Fig. 7 and Fig. 8 show lower Dice scores, which coincide with the regions closer to the edge of the brain. Figure 8 (Example 4), in particular, shows that the lack of detail in the images can result in widespread segmentation, increasing the region and decreasing the network's ability to locate fine details.

4. Discussion

Our research significantly advances glioma segmentation through the introduction of novel convolutional neural network (CNN) architectures, demonstrating promising results in both multiclass and

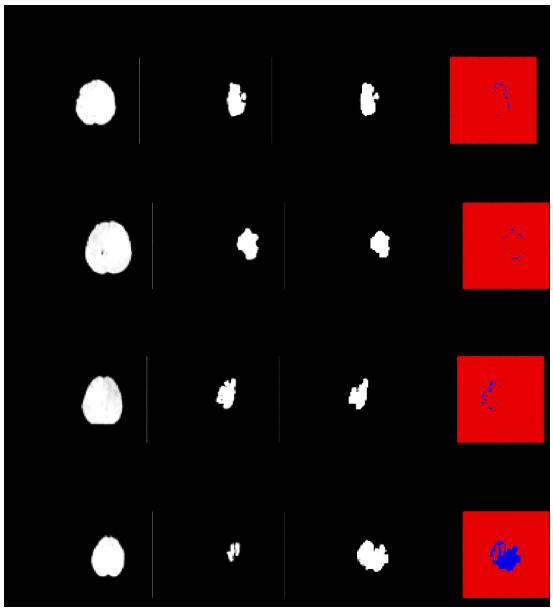


Fig. 8. Examples of architecture B test result.

anomaly segmentation tasks. Compared to existing solutions, our models offer an improved balance between accuracy and computational efficiency, crucial for potential clinical application. Specifically, Architecture B's near 90% success rate in anomaly segmentation underscores the effectiveness of CNNs in medical image analysis, aligning with prior studies that highlight deep learning's capability in complex segmentation challenges.

Where this study suggests that CNNs can elevate glioma image segmentation to a level comparable with professional radiologists, indicating a substantial gain in efficiency. After training, CNNs can process images much faster than manual segmentation methods. Furthermore, our research achieved segmentation performance on par with commercially available technologies, presenting CNNs as both accurate and efficient solutions. However, a detailed comparison of computational efficiencies remains unexplored in our report.

Theoretically, our findings enrich the discourse on customizing CNNs for the specific demands of medical imaging segmentation, demonstrating how these networks can be fine-tuned to effectively identify and delineate gliomas from MR images. Practically, integrating our CNN models into existing diagnostic procedures marks a significant leap in efficiency and precision, automating the segmentation process to lessen dependence on labor-intensive and potentially subjective manual analysis. This automation not only accelerates diagnostics but also enhances the accuracy of treatment planning, facilitating quicker, patient-specific therapeutic strategies.

A limitation of our study is the necessary downscaling of image resolution for processing, potentially affecting our models' ability to perform detailed segmentation. Future research should aim to overcome this challenge, leveraging more powerful computing resources or improved algorithms capable of processing high-resolution images without sacrificing performance.

This research contributes valuable insights into the application of deep learning models for brain tumor segmentation, providing a comprehensive comparison of different architectures and methodologies. Our findings are pivotal for enhancing tumor segmentation's accuracy and efficiency, with broader implications

for clinical diagnosis and treatment planning. By integrating advanced machine learning techniques, we deepen our understanding of tumor characteristics and their variability across patients, paving the way for more personalized and effective treatment approaches.

Addressing gaps in the literature, our work focuses on developing robust models capable of handling the complexity and diversity of brain tumor images. Our analysis underscores the importance of selecting a model that aligns with specific clinical requirements and dataset characteristics, considering the trade-off between model complexity, computational speed, and segmentation accuracy.

Furthermore, the study marks significant progress in automated medical imaging analysis, especially in neuro-oncology, by proposing a CNN-based method for segmenting gliomas using multimodal MRI data. This methodology enables glioma and its components' segmentation with accuracy comparable to that of professional radiologists and widely used commercial technologies, potentially leading to more precise and faster diagnoses, thus facilitating timely and effective treatment for brain cancers. The implications of this research are manifold, including improved diagnostic accuracy, enhanced clinical efficiency, personalized treatment planning, and fostering further research on automated systems for various malignancies.

Integrating the CNN model into clinical workflows involves ensuring compatibility with existing medical imaging and electronic health records systems, training for healthcare professionals to effectively use and interpret the model's results, and adhering to regulatory standards for clinical application. However, potential barriers such as technical challenges, resistance from healthcare workers, financial constraints, and regulatory and ethical considerations could hinder adoption.

Finally, this research underscores the importance of synergizing computational innovation with clinical practice, advancing AI's role in oncology for better patient outcomes and more efficient cancer treatments. Future research should focus on leveraging multimodal imaging data to improve tumor segmentation's precision and reliability and exploring the adaptability of our models to other brain tumor types, enhancing their applicability and impact.

5. Conclusion

A study was carried out on processing biomedical images through paradigms that surround electrical engineering, such as machine learning and signal processing. For this, the problem of detecting brain cancer from medical images such as magnetic resonance, a relevant problem in modern medicine, was presented. Deep learning algorithms based on convolutional neural networks were proposed to address this problem, and two deep neural network architectures were implemented and trained.

Architectures A and B were developed to target brain regions containing brain tumors and/or oedema. Architecture A was trained to locate not only the pixels of the images that contain or do not contain these anomalies, but also to characterize the type of anomaly among 5 classes: cancer nucleus, cancer affected region, oedema affected region and healthy region. An extra architecture (B) was trained only with the data of where the healthy region is, allowing the segmentation of anomalies in the brain.

The trained neural network architectures obtained promising results. Still, the need to reduce the resolution of the images used in training may have compromised the performance of the networks, leading to high values of the standard deviation of the results, in addition to segmentation results with a Dice score below 50% average for multiclass segmentation (architecture A). For the segmentation of anomalies, that is, of only one class (architecture B), the results were very positive, reaching close to a 90% hit rate.

With these results, it can be seen that the task of segmenting only one class is simpler than that of multiclass segmentation. It can be seen that network B can easily locate brain damage, but the finer details

tend to be merged with the region as a whole. This means that this network would be better used only for localization applications and detecting brain anomalies but not for fine segmentation.

In several cases of the image with less texture and contrast, the region affected by the tumor was included in the segmentation, but a region surrounding it was also included, suggesting that the network needs greater contrast and texture to be able to segment details finely. This situation may have been amplified by reducing the resolution of the original images, which directly results in a decrease in the fine details of the images.

There are several improvements proposed for possible continued studies of this study. Some of them are using the transfer learning concept, avoiding the need to train networks "from scratch,"; the using more powerful computers to make down-sampling the original images unnecessary, proposing deeper architectures, and making them capable of learning more complex features etc.

In conclusion, this study is intriguing in demonstrating that computational techniques based on the concepts learned during the undergraduate electrical engineering course are more than enough to perform tasks of high complexity and relevance in areas such as medicine and biology, so far removed from those traditionally addressed in this study.

Ethical approval

The College of Medicine's Committee of Shandong Provincial Third Hospital granted ethical approval for the study (KYLL-2020024). The participants have been provided with a clear understanding of the research objectives and voluntarily consented to participate in the study.

Data availability

Data for this study are available from the corresponding author upon reasonable request.

Funding

This study was self-funded by authors and did not receive any external funding.

Conflict of interest

The authors declare that there is no conflict of interest regarding the publication of this paper.

Author contributions

Jing Wang and Liang Yin participated in the design of this study. Liang Yin performed the statistical analysis. Jing Wang drafted the manuscript. All authors read and approved the final manuscript.

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